



Propagator

The Monthly Newsletter of the South Orange County Amateur Radio Association

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General Meeting

NPM Center
Apr 20, 2026
6:30 PM
Norman P. Murray
Center
23492 Veterans
Way, Mission Viejo,
CA 92692

President's Message

It is officially Spring and the flowers in my area are starting to bloom. My Jacaranda tree is showing signs of the beautiful, but messy purple flowers it has in abundance every year.

Although we don't have severe weather here in Southern California, there are still wind storms that cause damage to outside antennas and other items. It is a good time to survey your outside antennas and feed lines for damage and make repairs while the weather is nice and not as hot as it will be later in the summer. It's also a good time to grow some new antennas. Maybe a fan dipole, an End Fed Half Wave (EFHW) , Loop on Ground (LOG) or vertical?

Spring is a great time to take your gear portable and perhaps activate one of our many local parks. When was the last time you operated portable?

At our last SOARA meeting, the Board presented the following slate of officers and directors to the membership in attendance:

President	Dale Tyler (W6EDT)
Vice President	Greg Unruh (WE4BY)
Secretary	Charles Schultz (NY6I)
Treasurer	Ron Mosher (K0PGE)
Director at Large	Dan Upchurch (N6RPR)

There were three open Director at Large positions open for election this cycle.

Nominations from the floor were opened and there being no additional nominations from the floor, nominations were closed. Because none of

the positions up for election to the Board had more than one nominee, all of the persons nominated were deemed elected, per our new Bylaws that you recently adopted (Section VIII, subsection C). As such, no election ballots were sent out.

I was and still am hoping for more SOARA members to step up and volunteer for seats on the Board. We have quite a number of vacant positions. Your club needs help if we are to continue operating long term. Remember, the Director at Large positions are a great way to see what role you might best be suited for to help the club. Please send me an email or give me a call if you would like to discuss how you might join the Board.

Field Day is coming up Friday, June 26 through Sunday, June 28. We are looking for ideas on new antenna configurations for the 40M and 80M dipoles we have used in past years. The prior methods of using the ball field light standards no longer work well for several reasons, so we need to find new ways to support the antennas. Bring your ideas to the April or May meetings or to Soara Elmer Saturday on April 18 in Gilleran Park at 9:00 am.

73,

Dale Tyler - W6EDT

Membership Meeting

SOARA Membership Meeting – Polar Modulation

Join us for the SOARA monthly meeting on **Monday, April 20. As usual, we will gather at 6:30 PM, and start the main presentation about 7:00.**

FlexRadio has recently introduced its Aurora radio, which achieves very high efficiency through the use of polar modulation. QRP Labs has also used polar modulation to transmit SSB signals from its low power QMX/QMX+ radios. This technique avoids the tradeoffs between efficiency and linearity that are usually made in traditional transmitters and amplifiers.

I will attempt to explain what polar modulation is, and how it can be used in radio transmitters and amplifiers.

We are not planning to add Zoom for this meeting, so please show up in person if you want to participate.

Hope to see you there.

73, Greg Unruh, WE4BY

SOARA Saturday Report

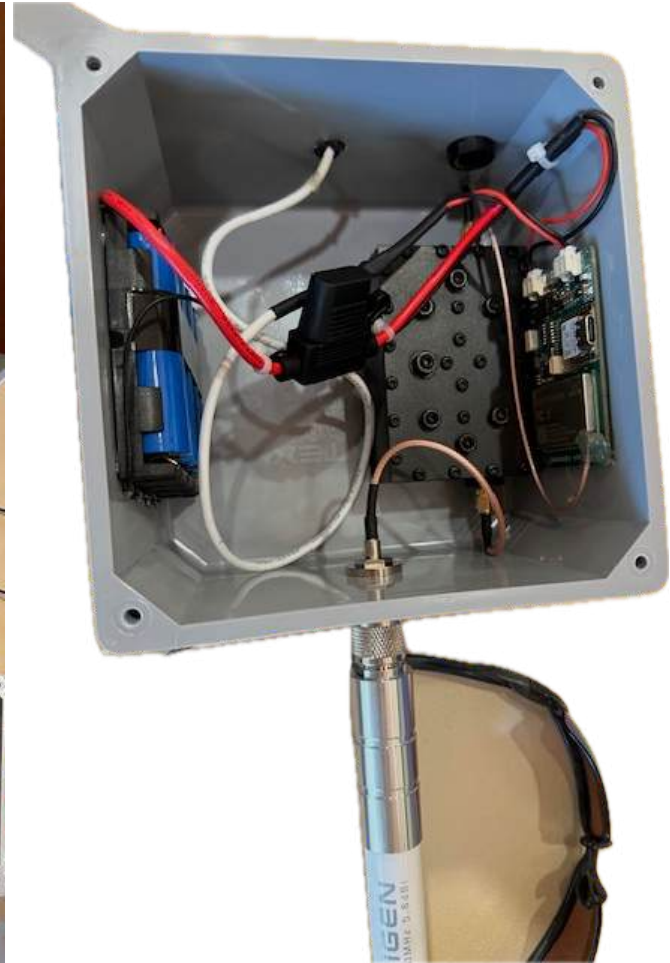
We had an interesting SES. We were at the Murray center. Greg brought donuts. We took in enough to pay Greg \$20 for the donuts and a \$1 balance for April. There were several generous donations. Thank you everyone!

Attendees were: W6TRG Tom, K6NOV Ray, KO6JCA Jeff, KC6FCY Charlie, NY6I Charles, KK6UMB Tom, N6PM Greg KO6CZG Mary, N6RPR Dan, W6CAG Claude, W6EDT Dale, AJ6LU Harald, WA6ED Ed, AD6OH Patti.

Charlie KC6FZY was testing an inductor with the nanovna. This inductor was for his HF antenna. In the photo below, the inductor is partially hidden by the mug. The black and red banana plugs help identify it.



Others demoed different aspects of meshcore. Dan N6RPR demoed a meshcore router. This included the outdoor antenna, batteries, waterproof enclosure, solar panel, a filter to block HF RF (black enclosure with lots of screws), and the CPU & transmitter. Note RTV on the U.FL connector to keep in position.



Ray K6NOV showed a meshcore tag. Not sure which brand. And other stuff.



Ed demoed Heltec ESP32 v3 kits in 3D printed cases. V3 is about 0.16 watt and seems to work okay in this area. Be sure to check other boards based on ARM processors like the T114. There is also a higher powered Heltec ESP32 V4 with more RF power. About 0.6 watts. The max allowable power is 1 watt per FCC. GPS is nice but not really needed for repeaters unless they are mobile. Goto meshcore flasher webpage to see what hardware is supported. Rak wireless also makes a board supporting 1 watt. There are probably more options.

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Send e-mail to membership@soara.org
Facebook: K6SOA Twitter: @K6SOA

Dale and Claude checking out the demos.



The April SOARA Elmer Saturday will be April 18, 2026 at the Murray Center. 9:00 am to noon. We will have various demos. Free CAD demo probably won't happen unless I get a volunteer. Ham radio and related demos are always welcome and appreciated.

73,

Ed WA6ED

Recommend Links

A few links to check out from the last month.

Hamclock is Back!! So someone picked up the hamclock project, check it out here: <https://www.youtube.com/watch?v=dwW-2iccWww>

Jeff, KO6JCA, shared this Meshtastic vs. MeshCore presentation from Yuma Hamfest: <https://www.youtube.com/watch?v=SZALDbLvLjI&t=3s>

Want to know what to take on a SOTA activation, check out the Ham Ninja's blog on this topic: <https://www.n1clc.com/2017/12/my-sota-expedition-loadout.html>

Ray, K6NOV

High-Gain Yagi (Meshcore US compatible)

1. Design Rationale & Theory of Operation

The primary objective was to construct a high-gain, 50 Ohm resonant antenna using common materials while maintaining professional-grade Return Loss (>30 dB).

- **Beta (Hairpin) Match:** A shunt-inductive matching system transforms the naturally low impedance of a Yagi driven element up to 50 Ohms. This provides a DC-grounded structure that is less sensitive to environmental static than a simple dipole.
- **The "Weather Margin" Philosophy:** The antenna was bench-tuned to a "super-null" at 922 MHz (-43 dB Return Loss). This 11.5 MHz upward offset acts as a deterministic buffer; environmental factors like humidity and PETG dielectric loading will pull the resonance downward into the target 910.525 MHz range upon deployment.
- **Air-Gap Isolation:** Suspending the elements 5 mm above the boom minimizes the dielectric influence of the wooden dowel, ensuring the SWR remains stable regardless of the wood's internal moisture content.

Wide Band Performance and Operational Latitude

The decision to utilize 14 AWG solid copper in conjunction with a Beta (Hairpin) Match provides this Yagi with a significantly lower Q factor than designs using thinner wire or traditional Gamma matches. In RF engineering, a lower Q results in an increased fractional bandwidth, which offers several deterministic advantages for this build:

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- **Impedance Stability:** The antenna maintains a usable SWR (under 1.5:1) across a broad range, centered at 922 MHz. This ensures that the 902–928 MHz ISM band is fully covered with high efficiency, even if the antenna undergoes minor physical shifting.
- **Dimensional Tolerance:** While the 12" caliper precision was critical for achieving the -43 dB "super-null," the wide-band nature of the design ensures that the antenna remains high-performing even if environmental factors (ice, dust, or UV-induced wood expansion) cause sub-millimeter changes in element spacing.
- **Reactance Compensation:** The hairpin match acts as a balanced shunt inductor that effectively "smears" the resonance point. This creates a "flat" SWR floor rather than a sharp, narrow "V" curve, making the antenna remarkably resilient to the dielectric loading of the PETG clips and the 5 mm air-gap standoff.

By bench-tuning to the higher end of the band, the design leverages this wide bandwidth to ensure that as the antenna "ages" or encounters humidity, the resonant point remains within the high-performance window of the target 910.525 MHz frequency.

2. Technical Specifications & Dimensional Logic

Frequency: 910.525 MHz

Element Material: 14 AWG Solid Copper (\$\varnothing\$ 1.63 mm)

Boom Material: 3/8" (9.5 mm) Wooden Dowel

Dimensional Coordinate Map

All measurements are Center-to-Center (C-C) referenced from the Reflector (0.0 mm).

Element	Total Element Length	Spacing (C-C)	Cumulative Position
Reflector	158.0 mm	—	0.0 mm
Driven Element	145.4 mm	49.4 mm	49.4 mm
Director 1	141.6 mm	65.8 mm	115.2 mm
Director 2	138.8 mm	78.1 mm	193.3 mm
Director 3	136.2 mm	84.2 mm	277.5 mm

Engineering Note on Radius Compensation:

To achieve a precise 6.0 mm C-C spacing, the physical wire ends at the feedpoint were cut to account for the 0.815 mm radius of the 14 AWG wire. A 4.4 mm (11/64") mandrel was used for the hairpin bend to ensure the final center-line spacing remained exactly 6.0 mm.

You can create the driven element utilizing individual wires soldered together, however I chose to manufacture the hairpin and driven element as a single piece.

Continuous-Wire Method: The Driven Element and Hairpin were fabricated from a single length of 14 AWG wire. The wire was center-bent to form the 20.0 mm deep "U" hairpin before transitioning into the radiator legs. This eliminates high-current solder joints. The PETG boom clips provide an opening wide enough for the hairpin.

Be sure to use a single radius for the hairpin bend and sharp 90° bends to start the driven element. This sharp bend will be the solder point for the coax.

3. Assembly & Fabrication Notes

Precision Metrology

- **Measurement:** A 12" digital caliper was used for all measurements. A 1.6mm error equates to a 10 MHz frequency shift at this band.
- **Preparation:**
 - Elements were cut slightly oversized, then filed square at the ends to reach final verified Center to Center dimension.
 - Element wire can be pressed into the clips with a flat-blade screwdriver.
 - Ensure each element is properly centered in the clip
 - Use the minimum of solder at the feedpoint to reduce capacitive effects of a solder "blob."
 - Small dots of urethane hot glue were applied sparingly to secure elements into the PETG clips and to lock the clips to the dowel once tuned. Epoxy may be a more suitable solution.
 - Positions were marked on the dowel using calipers and a fine pencil to eliminate additive errors.
 - Solder joints and open end of the coax are protected with Neutral-Cure (non-acidic) RTV to prevent copper corrosion.

Feedpoint Construction

Continuous-Wire Method: The Driven Element and Hairpin were fabricated from a single length of 14 AWG wire. The wire was center-bent to form the 20.0 mm deep "U" hairpin before transitioning into the radiator legs. This eliminates high-current solder joints. The PETG boom clips provide an opening wide enough for the hairpin.

Direct Feed & Choking: Coax was soldered directly to the feed point to minimize parasitic inductance. A Snap-on Ferrite Bead was installed immediately at the feed point to suppress common-mode currents.

Bonding & Weatherproofing

Boom Sealant: The dowel was treated with Clear Spar Urethane (Gloss). This marine-grade finish provides UV protection and a hydrophobic surface while remaining RF transparent. Other options for boom material

While the 3/8" wooden dowel is an excellent, low-dielectric choice for prototyping, moving to other materials requires balancing structural rigidity against the "velocity factor" and parasitic capacitance they introduce to the RF field. A preferred boom for a much more stable build would be fiberglass.

Fiberglass (G10/FR4) Rods

Fiberglass is arguably the best all-around material for a 900 MHz Yagi boom.

Fiberglass driveway marker stakes are available at home building supply stores and Amazon, however they are usually 5/16" (.31") diameter.

- **Pros:** It is completely non-conductive, has a very low dielectric constant (approx 4.0 to 5.0), and is impervious to moisture and UV. It will not warp or "rot" like wood.
- **Cons:** It is more expensive and can be difficult to bond to (hot glue does not adhere well to the smooth surface; two-part epoxy is required).

RF Impact: Nearly identical to wood, but with even better long-term stability in the -33 dB null.

Fiberglass driveway marker stakes are available at home building supply stores and Amazon. They would be an excellent choice for this build. I used what I had on hand (3/8 wooden dowel), but will probably pick up one up of these on my next trip to Home Depot. I adjusted a version of the clip to fit the 5/16" (.31") diameter.



[+more options](#)

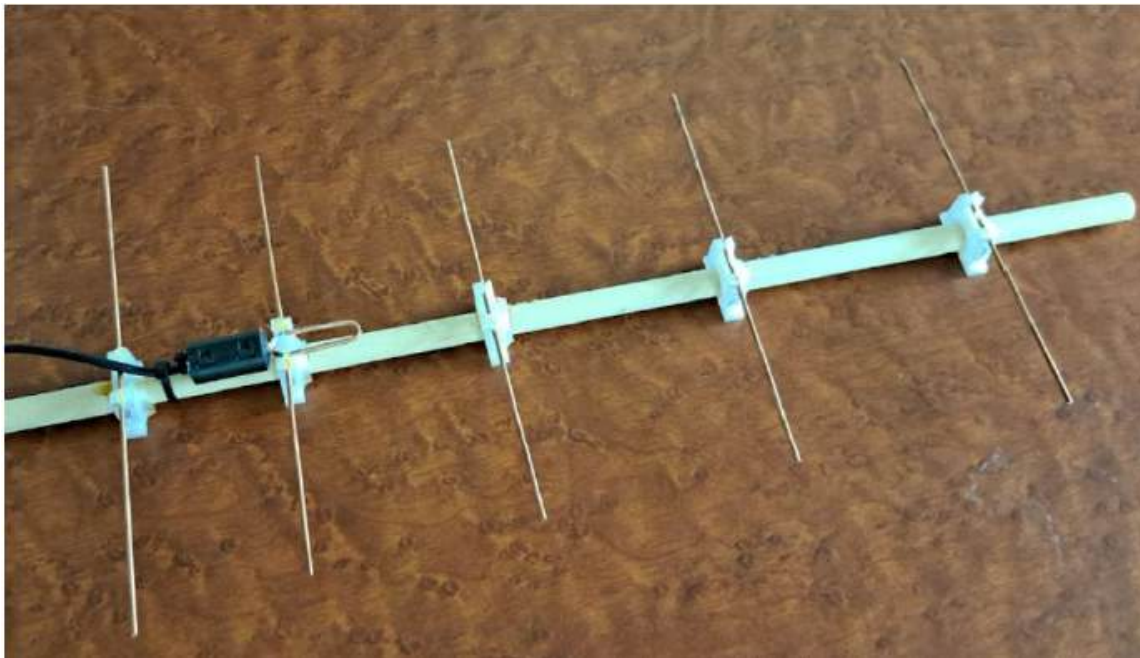
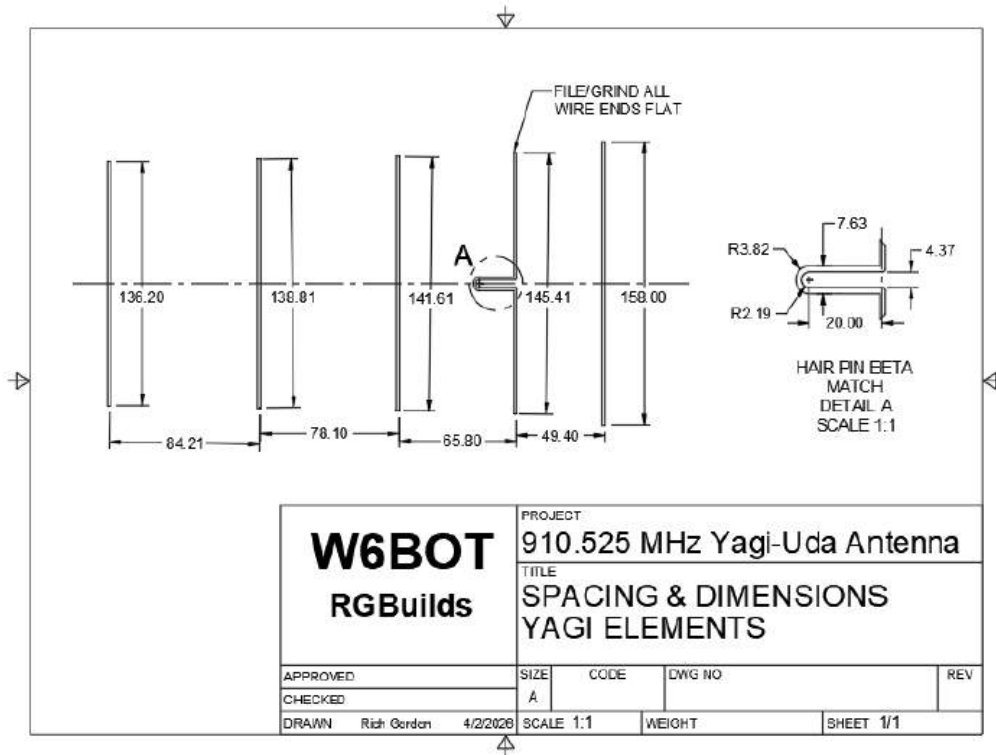
Everbilt 48 in. Orange Reflective Driveway Marker

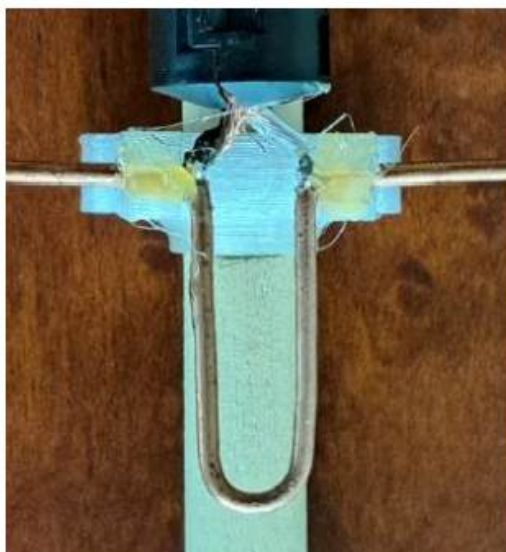
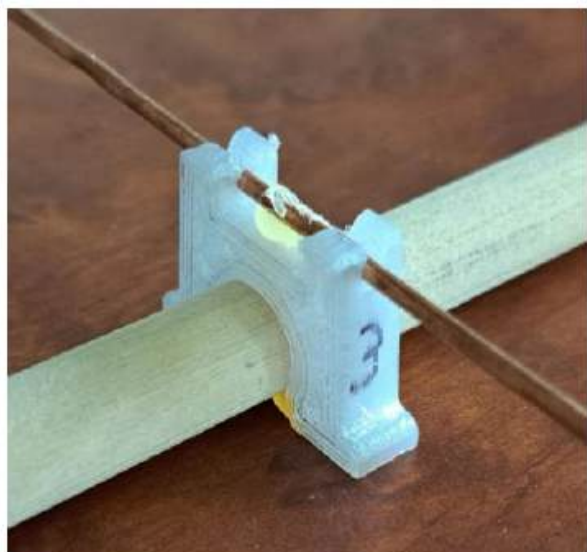
★★★★★ (4.7 / 933)
Model# 31474

\$3.47

4. Final Verification Logic

- **Sum of Parts:** 69.7mm (Leg A center-to-tip) + 6.0mm (Gap C-C) + 69.7mm (Leg B center-to-tip) = 145.4mm.
- **Bench Performance:** Verified Return Loss of -33 dB to -43 dB.
- **Stability Check:** Touching the element tips causes a measurable SWR shift, while moving the coax has no effect, confirming a properly isolated and balanced system.





Rich, W6BOT

SOARA - Statement of Cash Income & Expenses **October 1, 2025 Through March 31, 2026**

SOARA - Statement of Cash Income & Expenses			
October 2025 Through March 2026			
Six Months:	March	March	Increase
	<u>2026</u>	<u>2025</u>	<u>(Decrease)</u>
Cash Income:			
Memberships	7,735	9,418	(1,683)
Other & Interest	<u>2,076</u>	<u>1,751</u>	<u>325</u>
Total Cash Income	<u>9,811</u>	<u>11,169</u>	<u>(1,358)</u>
Cash Expenses:			
Repeaters including site rental, utilities & insurance	3,992	4,014	(22)
Member Activities	2,833	2,174	659
Other expenses	<u>2,661</u>	<u>1,035</u>	<u>1,626</u>
Total Cash Expenses	<u>9,486</u>	<u>7,223</u>	<u>2,263</u>
Cash Net Income	325	3,946	(3,621)
Beginning Cash-October 1	<u>28,179</u>	<u>25,745</u>	<u>2,434</u>
Ending Cash- March 31	<u>28,504</u>	<u>29,691</u>	<u>(1,187)</u>
Note: If any member has questions about the financials, please contact the Treasurer or any Board member.			

Ron Mosher - K0PGE, SOARA Treasurer

Calendars

Upcoming SOARA Events

Dates are subject to change. Check the SOARA Web Site (<http://www.soara.org>) to verify locations and times

This Month...

- | | |
|----------------------|--|
| April 18 9 AM – Noon | - SOARA Elmer Saturday, Gilleran Park |
| April 20 6:00 PM | - HAM Radio License Exams, Murray Center |
| April 20 6:30 PM | - General Meeting, Murray Center |
| April 27 7:00 PM | - Board Meeting, Murray Center |

Next Month...

- | | |
|--------------------|--|
| May 16 9 AM – Noon | - SOARA Elmer Saturday, Gilleran Park |
| May 18 6:00 PM | - HAM Radio License Exams, Murray Center |
| May 18 6:30 PM | - General Meeting, Murray Center |
| May 25 7:00 PM | - Board Meeting, Murray Center |

SOARA Information

SOARA meets at the Norman P. Murray Center, 24932 Veterans Way, Mission Viejo, CA on the third Monday of every month at 7:00 PM. For the months of January and February the third Monday is a holiday and the meeting is held on the fourth Monday.

License Exams: Amateur License Exams are given prior to SOARA meetings, except June. Exams are at 6pm. Prior registration is not required and walk-in applicants are welcome. For June, exams are held at Field Day. For further information, email Steve Kuver, K6UVR, at k6uvr@soara.org.

SOARA Library: SOARA has many amateur radio related books such as hand books, books about electrical theory, etc. available to lend out to club members. Contact library@soara.org for more info.

Web Site: SOARA maintains a web site with current club information. The URL is: <http://www.soara.org/>

Repeaters: The Laguna Beach, San Clemente, and Trabuco repeaters are open. The Santiago Peak repeaters are closed. For details or questions on the repeaters contact the repeater director, or repeater@soara.org.

2m	— 147.645	— (110.9)	Laguna Beach
2m	— 146.025	+ (110.9)	San Clemente
2m	— 145.240	— (110.9)	Trabuco
D-STAR 2m	— 146.115	+ (K6SOA C)	Laguna Beach
220	— 224.100	— (110.9)	Laguna Beach
220	— 224.640	— (pvt)	Santiago Peak. (C)
440	— 445.660	— (110.9)	Laguna Beach
D-STAR 440	— 445.705	— (K6SOA B)	Laguna Beach
440	— 447.180	— (pvt)	Santiago Peak. (C)
D-STAR 1.2G	1282.600	— (K6SOA A)	Laguna Beach

Nets:

- ☐ 40 meter HF (7.200 MHz +/- , Sundays @ 8 AM
- ☐ 10 meter HF (Technicians Welcome) (28.415 +/-) Sundays @ 9 AM
- ☐ General Membership Net - UHF/VHF (447.180, 147.645 & 224.640): Tuesdays @ 8 PM
- ☐ D-STAR (146.115 C module): Wednesdays @ 8 PM
- ☐ Tech Net - 147.645, 224.640, 447.180: Saturdays @ 9 AM
- ☐ California Rescue Communications (Gordo Net) HF (7.250 MHz +/- for QRM): Weekdays @ 8:30AM
- ☐ MVRACES - 447.180: Tuesdays @ 7PM
- ☐ Tri-Cities RACES - 146.025: Wednesdays @ 8 PM
- ☐ LNACS - 147.645: Thursdays @ 7 PM
- ☐ OC Parks Fire Watch - 447.180: Thursdays @ 8 PM

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